

10 Java GUI components

10.1 GUI design

A GUI interface is not designed casually. The designer has to pay close attention to something called “Look and Feel.” The guiding principle is making the GUI interface intuitive. To become technically proficient with the important “Swing” API library, you must become comfortable with two base classes and their functions:

- Component - paint() and repaint(), for example, originate in class Component.
- Container - A Container is a collection of related Components.

Swing borrows AWT functionality for:

- graphics,
- fonts and
- layout managers.

Lightweight components are ultimately derived from the AWT’s Container class, which extends AWT’s Component class. Therefore: Everything in Swing is a Container, is a Component.

10.2 Model-View-Controller architecture

Lightweight Swing components hand off their look and feel to a User Interface [UI] delegate. You could say look equals painting the component. And feel equals handling its events.

UI delegates can be plugged into a component when the component is constructed, or any later time. Thus, the term: “Pluggable look and feel”. Swing’s pluggable look and feel is based on a component architecture called: Model-View-Controller. The MVC architecture is designed for applications that need to wear many hats.

- Model—used to describe a way to maintain the data and provide access to it via accessor methods.
- View—used to paint a visual representation of some or all of the model’s data.
- Controller—used to handle events.

Model stores data, gives a way to access and modify data, fire events to registered Views, so the Views can update themselves based on the change in the underlying data. View takes care of displaying the data. And Controller handle events. ActionListeners and MouseListeners are MVC controllers.